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Quiz 7

An inverted index is to be created over the following documents.

Doc 1: "The quick brown fox jumps over the lazy dog near the riverbank."

Doc 2: "Near the peaceful river, the lazy cat watches the jumping fish."

Doc 3: "A brown rabbit and a quick squirrel play under the lazy afternoon sun."

Doc 4: "Foxes and dogs gather by the river to watch the sunset over the lazy waters."

To make the inverted index, the text is first normalized and split into tokens. Here is the list of tokens, and the list of documents they appear in (some lists are omitted).

the: [1, 2, 3, 4] quick: [1, 3] brown: [1, 3] fox: [1] jumps: [1] over: [1, 4] lazy: dog: [1] near: [1, 2] riverbank: [1] peaceful: [2]	river: cat: [2] watches: [2] jumping: [2] fish: [2] a: rabbit: [3] and: squirrel: [3] play: [3] under: [3]	afternoon: sun: [3] foxes: [4] dogs: [4] gather: [4] by: [4] to: [4] watch: [4] sunset: [4] waters:
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1. [3 points] Give the list of documents for the following terms:

a. lazy:

d. and:

b. river:

e. afternoon:

c. a:

f. waters:

2. [2 points] Calculate the term frequency (frequency of each term in each document) of the following terms:

a. the:

c. a:

b. lazy:

d. river:

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3. [2 points] Calculate the TF-IDF score of each **document** returned for the query “**lazy** AND **river**”. You may refer to the following table of values to help you calculate scores.
4. [3 points] Answer the following questions regarding the time complexity of performing intersection and union of postings:
- What is the time complexity of intersecting¹ two arbitrary² postings?
 - What is the time complexity of intersecting two postings where the last item of one of the postings is lesser than the first item of the other? Also mention the data structure used to store the postings.
 - Can you provide a sublinear bound³ on the time complexity of intersecting two postings where one of the postings is logarithmic in length to the other? Justify your answer.

¹ considering an optimal algorithm

² postings are always sorted

³ any polynomial less than n

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